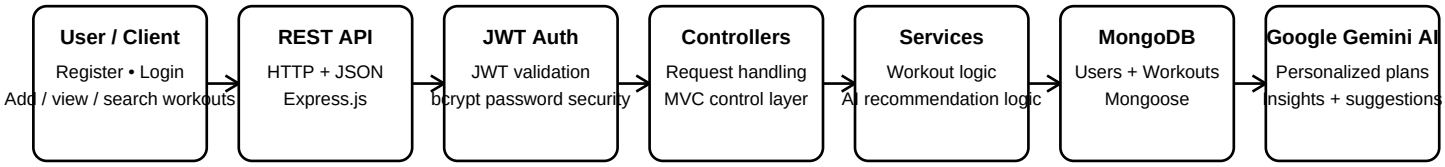
FITSENSE – AI-POWERED FITNESS RECOMMENDATION SYSTEM

SOLUTION ARCHITECTURE

Team ID: SWTID-2026-2908

Project Name: FitSense – AI-Powered Fitness Recommendation System

Architecture Flow:



Data & Response Flow

- 1. User sends a request through the REST API.
- 2. JWT authentication protects registered-user operations; passwords are secured with bcrypt.js.
- 3. Express routes direct requests to MVC controllers and service logic.
- 4. Workout data is stored and retrieved from MongoDB using Mongoose.
- 5. For recommendations and insights, user profile and workout statistics are sent to Google Gemini.
- 6. Gemini returns personalized fitness guidance; the service prepares the JSON response for the user.

Core Technology

Node.js • Express.js • MongoDB • Mongoose • Google Gemini • JWT • bcrypt.js • REST • JSON • MVC • Postman

Future Extension Points

Nutrition tracking • BMI calculation • Wearable integration • Workout reminders • Fitness analytics dashboard